■ MORUS-LENS

Mulberry Leaf Pathology Intelligence & Sericulture Agronomic Portal

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Executive Summary:
MORUS-LENS is an intelligent agricultural computer vision and decision-support portal built specifically for sericulture (silk farming) cultivators, moriculture agronomists, and pathology researchers. The system enables a farmer to capture a mulberry leaf (*Morus alba / Morus indica*) photo using any smartphone. Within one second, the platform isolates the leaf blade from surrounding clutter, accurately diagnoses whether the leaf is **Disease Free**, afflicted with **Leaf Rust**, or afflicted with **Leaf Spot**, and provides a clinical treatment recipe along with a mandatory silkworm chemical feed-withholding calendar to prevent lethal silkworm poisoning.

1. Background & The Real-World Agricultural Problem

What is Sericulture and Why is Mulberry Sacred?
Sericulture is the agro-industry that produces raw silk, supporting millions of rural livelihoods across India (notably in Tamil Nadu, Karnataka, and Andhra Pradesh) and Asia. The sole engine of this industry is the domestic silkworm, *Bombyx mori*. Silkworms are **strictly monophagous**—meaning they eat exclusively fresh, tender leaves of the mulberry tree (*Morus alba* or *Morus indica*) and cannot survive on any other food source.

The nutritional chemistry and moisture of the mulberry leaf directly dictates the silk gland development of the larva, the cocoon shell weight, filament reelability, and resistance to catastrophic silkworm ailments such as *flacherie* and *grasserie*.

The Major Foliar Pathologies:

- **Mulberry Leaf Rust (*Cerotelium fici*):** Erupts into thousands of brownish-orange powdery spore mounds (uredinia) on the lower surface of the leaf blade. Causes rapid yellowing, severe dehydration, and premature leaf drop.
- **Mulberry Leaf Spot (*Pseudocercospora mori*):** Forms irregular dark brown necrotic lesions surrounded by yellow chlorotic halos that dry up and fall out into 'shot-holes', depleting crude protein by up to 40%.

The Deadly Hazard: The 'Pesticide Paradox'

When farmers notice fungal infections, they frequently spray chemical fungicides (e.g. Mancozeb, Carbendazim, Copper Oxychloride). However, chemical residues remain active on the foliage for days. If a farmer harvests these leaves too early and feeds them to larvae, **the silkworms suffer acute chemical poisoning and perish en masse within 24–48 hours**. Therefore, a diagnosis is useless without a strict **chemical feed-withholding calendar**. MORUS-LENS addresses this life-or-death challenge directly.

2. Exhaustive Folder Structure & File Purpose Breakdown

To ensure that anyone inspecting the codebase understands exactly why each component exists, the table below maps out every single file, its function, and its role in the application architecture:

File / Folder	Type & Location	Purpose & Operational Role in Project
app.py	Root Application	The core Flask server script. Coordinates URL routing, user authentication, SQLite database models, OpenCV Excess Green segmentation, neural network inference, and recommendation generation.
streamlit_app.py	Root Script	An alternative lightweight data science prototype allowing researchers to test the vision model rapidly without setting up user accounts.
train_advanced_model.py	Root Laboratory	The laboratory training script. Implements EfficientNetB0 transfer learning, heavy agricultural augmentations, cosine learning rate warmup, and class weighting.
mulberry_leaf_...keras	Model Weights (39.7 MB)	The pre-trained deep neural network weights storing millions of mathematical coefficients that recognize leaf textures.
requirements.txt	Configuration	Specifies all required Python libraries (Flask, SQLAlchemy, OpenCV, Pillow, reportlab, streamlit).

instance/ mulberry_app.db	Database	Relational SQLite database storing users (with PBKDF2 hashed passwords), scan histories, and inquiries submitted via contact desk.
static/css/ style.css	Design System	Complete CSS stylesheet defining emerald sericulture styling, responsive mobile drawer, animated progress bars, and badges.
static/uploads/	Asset Storage	Partitioned file archive storing uploaded leaf images isolated by user ID (static/uploads/<user_id>/).
templates/ base.html	Master Template	The structural skeleton for all pages. Contains top navigation bar, mobile hamburger menu, alerts, and footer.
templates/ dashboard.html	Main Screen	The diagnostic scanner screen with drag-drop, mobile camera shutter, live animated probability gauges, and printable report.
templates/ history.html	History Portal	User's farm diagnostic journal with filter tabs (All, Healthy, Rust, Spot, Unknown) and inspection modal pop-up.
templates/ guide.html	Botanical Handbook	Organic Moriculture Herbarium featuring a 4-step dichotomous diagnostic key, specimen profiles, and feeding calendar.
templates/ contact.html	Support Desk	Developer Sathish Kumar's contact hub with direct 1-tap WhatsApp (+91 86105 25967), email, LinkedIn, and inquiry ticketing.
templates/ login.html & register.html	Auth Portal	Secure sign-in and account registration pages. Legacy demo accounts are purged for strict private user data separation.

3. End-to-End System Architecture & Data Flow

The Lifecycle of an Image Diagnosis:
From the moment a farmer holds their smartphone up to a mulberry tree to the moment a diagnostic report appears, the request travels through 6 tightly coordinated phases:

1. **Image Acquisition:** The farmer clicks or selects an image on dashboard.html. Tapping the upload zone on a mobile device immediately opens the rear phone camera (capture='environment').
2. **Secure Transmission:** JavaScript sends an asynchronous multipart HTTP POST request to the /predict endpoint on the Flask server.
3. **Disk Isolation:** app.py validates user authentication via Flask-Login, assigns a secure UUID filename, and saves the file to static/uploads/<user_id>/.
4. **Botanical Computer Vision Segmentation:** The image is passed to OpenCV in analyze_botanical_pathology(). The Excess Green formula (2G - R - B) and morphological contour tracking isolate the green leaf blade while discarding hands, tables, and backgrounds.
5. **Neural Network Inference & OOD Gating:** The normalized tensor is evaluated by the neural network. If confidence is below 70% or the margin over the second-place class is below 20%, the Out-of-Distribution filter re-routes the result to **Unknown / Ambiguous**.
6. **Persistence & UI Rendering:** The diagnosis is saved in the SQLite predictions table and returned as JSON. The browser animates the probability bars, displays the prescription badge, and provides a PDF print button.

4. How the AI Works (Explained in Simple Terms)

For non-technical readers, the Artificial Intelligence inside MORUS-LENS works in three simple steps:

Step A: The 'Sunglasses' Filter (Isolating the Leaf)

When a farmer places a leaf on a wooden table or holds it with fingers, ordinary AI programs get confused by the skin or wood colors. MORUS-LENS applies mathematical 'sunglasses' called the **Excess Green Index (2G - R - B)**. Living chlorophyll reflects massive green light and absorbs red and blue light. By calculating 2 × Green - Red - Blue, the plant leaf turns glowing white, while tables, fingers, and shadows turn black. The system draws a boundary around the leaf and ignores everything outside it.

Step B: The 'Magnifying Glass' (Examining Symptoms)

Inside the isolated leaf, the computer measures the color spectrum. Fungal Rust produces warm amber-orange pustules in HSV color space, while Leaf Spot produces dark sunken dead tissue (necrosis) surrounded by chlorotic yellow halos. Healthy areas display uniform, bright chlorophyll.

Step C: The Deep Learning Brain (The Specialist Doctor)

Finally, the image enters a Convolutional Neural Network (EfficientNetB0). The network acts like layers of specialist detectives: early layers check veins and edges; middle layers look for spore circles; deep layers recognize complex disease signatures learned from thousands of mulberry samples.

5. Pathological Classes & Silkworm Feeding Withholding Calendar

Clinical Condition	Typical Symptoms	Agronomic Control	Silkworm Feeding Safety Window
■ **Disease Free** *(Healthy Morus)*	Smooth vibrant green blade, intact veins, glossy waxy cuticle.	Split NPK (300:120:120 kg/ha/yr) + 20 tons Farm Yard Manure.	**0 Days — Safe for immediate feeding**
■■ **Potential Leaf Rust** *(Cerotelium fici)*	Brownish-orange powdery spore mounds (uredinia) on underside.	Spray Mancozeb 0.2% (2g/L) or Copper Oxychloride 0.2% (2g/L).	**10–14 Days Withholding DO NOT FEED SPRAYED LEAVES!**
■ **Potential Leaf Spot** *(Pseudocercospora mori)*	Dark brown necrotic spots with yellow halos; shot-holes.	Spray Bavistin (Carbendazim 0.1%, 1g/L) or Chlorothalonil 0.2%.	**10–12 Days Withholding DO NOT FEED SPRAYED LEAVES!**
■ **Unknown / Ambiguous** *(Out-of-Distribution)*	Blurry image, non-mulberry leaf, or novel ambiguous pathogen.	Retake in daylight. Consult Sericulture Extension Officer.	**Quarantine foliage until confirmed**

6. Key System Upgrades & How to Run the Application

Major Upgrades Engineered by Sathish Kumar:

1. **Project Rebranding:** Renamed all views, templates, navigation headers, and documentation to **MORUS-LENS**.
2. **Demo Account Purge:** Removed all pre-seeded demo accounts from database startup and templates, enforcing true multi-user privacy.
3. **Mobile Responsiveness:** Built a collapsible hamburger drawer, fluid touch targets, and smartphone camera shutter integration.
4. **Organic Botanical Field Handbook:** Replaced artificial blocks with an organic herbarium featuring dichotomous diagnostic keys.
5. **Dedicated Contact Desk:** Integrated Sathish Kumar's profile with direct 1-tap WhatsApp (+91 86105 25967), email, LinkedIn, and SQLite ticketing.
6. **Botanical Hull Segmentation:** Implemented Excess Green OpenCV morphology to eliminate false positives from white tables and holding fingers.

Step-by-Step Installation & Execution Guide:

 # 1. Clone the repository and enter directory
git clone https://github.com/sathishkumar1612/Mulberry-Leaf-Disease-Prediction.git
cd Mulberry-Leaf-Disease-Prediction

 # 2. Activate Python Virtual Environment
myenv\Scripts\activate *(Windows)* | source myenv/bin/activate *(Linux/macOS)*

 # 3. Install Requirements & Start Server
pip install -r requirements.txt
python app.py

 # 4. Access the Web Application: Open browser at **http://127.0.0.1:5000**

Developer & Project Credits:
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MORUS-LENS — Empowering Sericulture Farmers with Lab-Grade Agricultural Artificial Intelligence.